

The empirical recurrence for OEIS A297759: recovering a conjecture that is not written down

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Abstract

OEIS A297759 records “Empirical recurrence of order 78”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 127$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 78, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 78 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A297759 is “Number of $n \times 5$ 0..1 arrays with every 1 horizontally, diagonally or antidiagonally adjacent to 2, 3 or 4 neighboring 1s.”. It has offset 1 and begins

1, 78, 734, 7081, 109228, 1456855, 18733087, 251641086, ...

The entry states:

Empirical recurrence of order 78 (see link above)

As of the “Last modified” line on the live entry (Jan 05 2018 18:58:21, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 127$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 127$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 254$ exact terms of the model returns order 78 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{78} c_i a(n-i),$$

c_1	6	c_{21}	1110115111540	c_{41}	65167555458533	c_{61}	−120046598584
c_2	32	c_{22}	2061464391656	c_{42}	51654159430912	c_{62}	1797296849337
c_3	945	c_{23}	920867854302	c_{43}	−10012893691269	c_{63}	351419566181
c_4	344	c_{24}	−4490328689622	c_{44}	−64152149675182	c_{64}	−665811791210
c_5	−3866	c_{25}	−9128960669282	c_{45}	−97897380090052	c_{65}	−88953257209
c_6	−184141	c_{26}	−5805084710194	c_{46}	−28302495242928	c_{66}	113463920382
c_7	−327144	c_{27}	11848258103908	c_{47}	56929126257689	c_{67}	35836554348
c_8	252890	c_{28}	26935287056242	c_{48}	79674152269342	c_{68}	−31005966903
c_9	14921664	c_{29}	19595073781481	c_{49}	20998461436607	c_{69}	2922148678
c_{10}	32447349	c_{30}	−23223535917552	c_{50}	−40844269775613	c_{70}	2659111951
c_{11}	−14051860	c_{31}	−56933990354535	c_{51}	−38984268043629	c_{71}	531769559
c_{12}	−601617811	c_{32}	−42001434746129	c_{52}	11118263200378	c_{72}	−749954425
c_{13}	−1302789485	c_{33}	34033683684419	c_{53}	32107795700738	c_{73}	219657207
c_{14}	368153380	c_{34}	92107422471192	c_{54}	−1262243920579	c_{74}	−15980671
c_{15}	13152162483	c_{35}	64120791737388	c_{55}	−28353354986671	c_{75}	−1470797
c_{16}	25951648111	c_{36}	−32863271998704	c_{56}	−11219142699394	c_{76}	−25129
c_{17}	−2115539809	c_{37}	−112885835867519	c_{57}	15751323448105	c_{77}	51558
c_{18}	−160845718272	c_{38}	−72929017808698	c_{58}	12779701614507	c_{78}	−3372
c_{19}	−294578486548	c_{39}	7632414416411	c_{59}	−2513276290321		
c_{20}	−52780047296	c_{40}	83832575560638	c_{60}	−6345069004231		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 78, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $78 + 4$ terms removed returns order 78 as well, so $\deg q_{\min} = 78 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 17 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $78 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 78 that this sequence satisfies — and that family turns out to have exactly one member.

References

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